

Kalyani Patrikar

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Building computational frameworks with neural networks, quantum chemical methods, physics-based models to uncover mechanisms, enhance system performance, and design new materials

EXPERIENCE

Postdoctoral Research Associate, *University of Liverpool*

PRESENT

- Developed **probabilistic model** for charge transport in mixed conducting polymers with **time variant disorder**
- Developing **physics-based model** for charge transfer at device interfaces

Early Career Fellow, *Indian Institute of Technology Gandhinagar*

Aug 2022 – Oct 2024

One of the highest postdoc fellowships offered by any Indian institute secured with independent proposal

- Designed **surrogate** general purpose **graph neural network** with accuracy in prediction for diverse quantum mechanical properties from structure, demonstrated efficiency of **attention networks**, accurate for small datasets, performed **conformal analysis**
- Developed **multiscale simulation pipeline** involving ab initio computations, semi empirical QM, and **statistical model**, predicting device parameters validated against experimental data; integrated **MLIP** derived **MD** conformers into charge transport simulation pipeline; fine tuned MACE on Y6-based configurations to enable rapid conformer prediction across the Y6 molecular family
- Developed** automated fragmentation **algorithm** to generate new molecules analysed with neural network, validated with DFT
- Proposed** the **mechanism** for contact resistance in organic devices, demonstrated by combining device characterisation with ab initio computations and **physics-based model**, extended for functionalised device interfaces

Senior Research Assistant, *Center for Excellence in Nanoelectronics*

Jul 2013– Dec 2014

- Designed and fabricated oxide nanoparticles embedded organic transistors for toxic gas sensors

SKILLS

ML Frameworks : GNN & GAT, Random Forest, PINN

Modeling : Probabilistic model, physics-based models, ab initio computations

Scientific Software : CP2K, Gaussian16, Quantum Espresso, xTB, GROMACS, Jmol, VMD

Programming : **Python**(PyTorch, PyG, scikit-learn, RDKit, ASE, MACE, NumPy), Bash (& SLURM)

KEY PUBLICATIONS (COMPLETE LIST AT kalyanipatrikar.github.io/publications)

Journal (*Equal Contribution Authors, † Corresponding Author)

- Dinesh Trivedi*, Kalyani Patrikar*, Anirban Mondal, "Dataset Size Invariant Graph Attention Network for Prediction of Diverse Molecular Properties", *Mol. Syst. Des. Eng.*, 9, 2024 [[10.1039/D4ME00113C](https://doi.org/10.1039/D4ME00113C)] **⟨ Surrogate GAT model, accurate for electronic as well as optical molecular properties, and for small datasets ⟩**
- Kalyani Patrikar, Keval Patadia, Rudranarayan Khatua, Anirban Mondal, "From Molecules to Devices: A Multiscale Approach to Evaluating Organic Photovoltaics" *J. Chem. Theory Comput.*, 20, 22, 2024 [[10.1021/acs.jctc.4c01029](https://doi.org/10.1021/acs.jctc.4c01029)] **⟨ Computational pipeline: molecule in device performance out ⟩**

3. Bibhas Das, Kalyani Patrikar, Atharva Keny, Anirban Mondal, "From fragments to function: data-driven design of high-performance non-fullerene acceptors for organic photovoltaics" *Mol. Syst. Des. Eng.*, 11, 269-291, 2026 [10.1039/D5ME00182J] < **Automated fragmentation and combinatorial generation of novel molecules** >
4. Kalyani Patrikar[†], Valipe Ramgopal Rao, Dinesh Kabra, Anirban Mondal[†], "Understanding the Microscopic Origin of the Contact Resistance at the Polymer–Electrode Interface", *ACS Appl. Mater. Interfaces*, 15 (42), 49427, 2023 [10.1021/acsami.3c10260] < **DFT and physics-based model with device characterisation to demonstrate mechanism of interface resistance in organic transistors** >
5. Kalyani Patrikar[†], Anirban Mondal[†], "Deciphering the Microscopic Phenomenon Behind Contact Resistances in Interlayer Functionalized Electrodes and Organic Semiconductors", *Physical Review Materials*, 8, 054606, 2024 [10.1103/PhysRevMaterials.8.054606] < **DFT and physics-based model for mechanism of charge transfer at functionalised interfaces** >

Conference (*Presenting Author)

1. K. Patrikar*, C. Burke, A. Troisi, "Charge Transport Enhancement with Dynamic Disorder", Oral presentation at *European Conference in Molecular Electronics 2025* < **Probabilistic model for charge transport in dynamically disordered systems** > Awarded **Best Contributed Talk**

EDUCATION

Doctor of Philosophy

Indian Institute of Technology Bombay

Electrical Engineering

Jan 2015 – Dec 2021

Master of Technology

Indian Institute of Technology Bombay

Materials Science

Jul 2011 – Jun 2013

Bachelor of Technology

Visvesvaraya National Institute of Technology

Metallurgical & Materials Engineering

Jul 2007 – May 2011

PHD THESIS

Bulk and Interfacial Charge Transport in Organic Field Effect Transistors

- Demonstrated **novel** charge doping **mechanism** of self-assembled monolayers at transistor contacts; developed **device model** based on ab initio computations and device characterisation
- Established new molecular design paradigm for order-of-magnitude **device performance** improvement based on device characterisation, ab initio computations, and **polaron transport** analysis
- Developed **multiscale model** simulating measured cryogenic device characteristics of organic transistors

TEACHING AND MENTORSHIP

- **Teaching:** Flexible Electronics (2 Credit Course) IIT Gandhinagar
- **Demonstrator:** Introduction to Spectroscopy, **Chemical Data Discovery and Design** University of Liverpool
- **Teaching Assistant:** Physics of Transistors, **Machine Learning**, Microelectronics Lab, Communication Skills, VLSI Technology, Thermodynamics IIT Bombay
- **Students Mentored:** B.Tech and MSc. students for thesis project work IIT Gandhinagar

EXTRACURRICULAR ACHIEVEMENTS

- Organised and anchored events as the Institute Executive Member in IIT Bombay Academic Council
- Completed high altitude treks in Himalayas, India